Fairoz Ahmad Sheikh

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Professional Summary

Python developer and AI/ML Engineer with experience building backend services, REST APIs, and data processing and automation solutions, alongside Machine Learning, Generative AI, LLM, and RAG applications. Strong foundation in Object Oriented Programming, SQL, and Data Structures & Algorithms, with hands-on experience across the software development life cycle, from coding and debugging to deployment using Docker and AWS. M.Tech in Artificial Intelligence .

Education

Islamic University of Science & Technology, Pulwama Aug 2024 – Jun 2026
Master of Technology (M.Tech) in Artificial Intelligence **CGPA: 9.1/10**

Government College of Engineering & Technology, Safapora 2018 – 2023
Bachelor of Technology (B.Tech) in Computer Engineering **CGPA: 7.5/10**

Professional Experience

AI/ML Engineer Intern — DevMatrix Dec 2025 – Aug 2026

- Developed Python-based applications and backend services for a UK-based fintech project covering financial data analysis, fraud detection, and process automation.
- Designed and developed REST APIs and backend services to expose Machine Learning and LLM-based functionality to applications.
- Built data processing and automation pipelines, including preprocessing, feature engineering, and reporting, to support analytics and compliance processes.
- Implemented Machine Learning and Deep Learning workflows for anomaly detection, predictive analytics, and data-driven insights.
- Built applications using Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and Graph RAG, with semantic search over embeddings and vector databases.
- Worked across the software development life cycle, including coding, debugging, evaluation, deployment, and performance optimization.
- Collaborated with cross-functional teams to understand requirements and deliver scalable solutions.

AI Data Trainer — Invisible Technologies Remote (Contract) Aug 2026 – Present

- Evaluate, correct, and rank model-generated Python and technical responses against correctness, reasoning quality, and instruction-following criteria for large language model training pipelines.
- Author reference solutions and structured critiques used as supervision data, applying rubric-driven review standards while maintaining a high task acceptance rate under strict quality review.

Technical Skills

- **Programming:** Python, Object Oriented Programming (OOP), Data Structures & Algorithms (DSA), SQL, JavaScript, HTML, CSS, C++
- **Web & Backend:** FastAPI, Flask, REST APIs, Pydantic, Streamlit
- **Databases:** SQL, MongoDB, Vector Databases (Qdrant, ChromaDB, Pinecone)
- **DevOps & Deployment:** Git, GitHub, Docker, AWS, DVC, Render, Vercel
- **Machine Learning:** Classification, Regression, Clustering, Anomaly Detection, Feature Engineering
- **Deep Learning:** CNNs, Transfer Learning, TensorFlow, PyTorch
- **NLP & Generative AI:** NLP, Text Classification, Semantic Search, LLMs, RAG, Graph RAG, Embeddings, Prompt Engineering
- **AI Frameworks & Tools:** LangChain, LlamaIndex, Hugging Face Transformers, OpenAI API, Gemini API
- **Data Science:** Pandas, NumPy, Scikit-learn, Matplotlib

Selected Projects

Insurance Premium Predictor – End-to-End ML Application *Python, Scikit-learn, FastAPI, Flask, Streamlit, Docker, AWS*

- Architected and deployed an end-to-end machine learning application for real-time insurance premium prediction.

- Developed a production-oriented FastAPI backend with model inference, Pydantic input validation, and structured JSON responses.
- Containerized the application using Docker and deployed the backend to AWS for consistent and scalable execution.
- Developed Flask and Streamlit interfaces for user interaction, visualization, and prediction workflows.
- Backend Repository

DocuMind AI – Retrieval-Augmented Document Question Answering *LangChain, OpenAI API, Pinecone, ChromaDB, Streamlit*

- Developed an intelligent RAG-based document question-answering system for extracting reliable insights from private documents.
- Implemented document ingestion, text chunking, embedding generation, semantic search, vector indexing, and LLM-based response generation.
- Built an interactive Streamlit interface supporting PDF, DOCX, and TXT document uploads with context-aware chat history.
- Optimized retrieval quality using configurable top-k retrieval and chunk-size tuning.
- GitHub Repository

NeuroScholar – Hybrid RAG System for Research Paper Comprehension *Python, LangGraph, RAG, Graph RAG, ChromaDB, NetworkX*

- Developed a privacy-preserving AI assistant for research paper comprehension combining Vector RAG and Graph RAG.
- Implemented semantic retrieval using sentence-transformer embeddings and ChromaDB for context-aware document search.
- Built a directed knowledge graph using NetworkX to represent entities and semantic relationships extracted from research papers.
- Designed a LangGraph-based pipeline for query routing, vector retrieval, graph retrieval, context merging, response generation, and iterative evaluation.
- Implemented an LLM-as-a-judge evaluation loop to assess generated responses and improve retrieval and generation quality.

DDoS Attack Detection Using Machine Learning *Python, Machine Learning, Cybersecurity*

- Developed a machine learning-based system for detecting and classifying DDoS attacks from network traffic data.
- Implemented data preprocessing, feature extraction, model training, evaluation, and prediction using supervised learning techniques.

Mini Food Vision – Food Image Classification *TensorFlow, CNN, Transfer Learning*

- Developed a CNN-based image classification model for 101 food categories using the Food-101 dataset.
- Applied transfer learning to improve classification performance and reduce training requirements.

Additional Projects

Fingerprint-Based Door Lock System *C++, Arduino, Embedded Systems*

- Built a biometric access-control system with an AS608 fingerprint sensor, implementing enrollment, authentication, and fail-secure relay control.

SkimLit – Medical NLP Research Reproduction *TensorFlow, NLP, Deep Learning*

- Reproduced a medical abstract classification study using sequential sentence classification models for biomedical text.

Certifications

- Python with Data Science & Machine Learning – IIT Jammu
- IBM Data Scientist Professional Certificate – Coursera
- Machine Learning – University of London, Coursera
- Full Stack Web Development Bootcamp – AlmaBetter

Achievements

- Ranked among the top cybersecurity learners in India on TryHackMe.
- Achieved Top 0.1% global ranking on the TryHackMe platform.
- Maintained a 500+ day coding streak on GitHub.

Research Interests

Deep Learning • Computer Vision • Natural Language Processing • Generative AI • Retrieval-Augmented Generation